

1 Induction to prove alternating sign on implies roots

Statement: $Q(n)$ if every adjacent pair of integers in the set $\{0, \dots, n\}$, when $n \geq 1$, alternate sign under f (e.g. $f(0) < 0 \wedge f(1) > 0$), then there is a set S that contains root of f , where $|S| = n - 1$. Formaly

$$Q(n): \forall l \in \{0, \dots, n\}: R(f, l) \implies \exists S \left((|S| = n) \wedge (\forall s \in S : f(s) = 0) \right)$$

Base Case: $n = 1$

$$Q(1): \forall l \in \{0, 1\}: R(f, l) \implies \exists S \left((|S| = 1) \wedge (\forall s \in S : f(s) = 0) \right)$$

Assume that for some f , the image of 0 and 1 under f is $f(0) \leq -\frac{1}{6}$ and $f(1) \geq \frac{1}{6}$ respectively. Then, by Intermediate value theorem, theres some $s_0 \in [0, 1]$ satisfying $f(s_0) = 0$.

Take $S = \{s_0\}$. Clearly, $|S| = 1$, which is what we wanted. This completes the base case.

Inductive case: Suppose $Q(k-1)$ is true for $k-1 \geq 2$. Then we must show it is also true $Q(k+1)$.

Use the inductive hypothesis to obtain the S_{k-1} where S satisfies

$$\left((|S| = 1) \wedge (\forall s \in S : f(s) = 0) \right)$$

We want to show there is a set S_k , where $S_{k-1} \subseteq S_k$ and $S_{k-1} \cap S_k =$, and where S_k satisfies $\left((|S| = 1) \wedge (\forall s \in S : f(s) = 0) \right)$.

Case 1: $k-1$ is even; moreover, by assumption $P(f, k-1)$ is true. This means that $k-1 \geq \frac{1}{6}$.

Similarly, because $k-1$ is even, k is odd. by assumption $P(f, k)$ is true. This means that $k \leq -\frac{1}{6}$.

Therefore, we have two points for which the range of f cross from negative to positive. By intermediate value thoerem, there must be a root, call it s_k

Observe that $\{s_k\} \cap S_{k-1} =$, since the intervals, $[0, 1] \dots [k-2, k-1]$, which produced S_{k-1} are strictly less then k.

Therefore take $S_k = \{s_k\} \cup S_{k-1}$.

Finally, note that for any two disjoint sets A, B , we have that $|A \cup B| = |A| + |B|$. So, $|S_k| = |\{s_k\} \cup S_{k-1}| = 1 + k - 1$.

This gives the claim, in the case that k is even.

Case 2: By similar argumentation made in case 1, this case also is true.

This completes the inductive hypothesis.

Conclusion: We have shown the claim to be true.